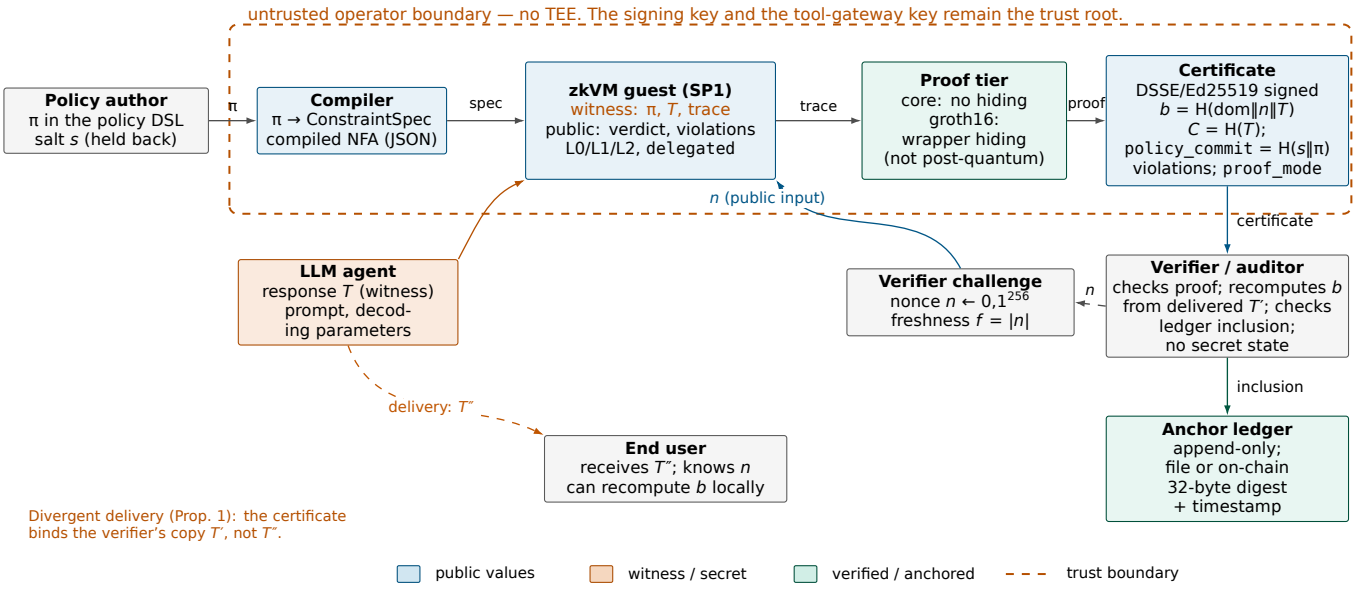
untrusted operator boundary — no TEE. The signing key and the tool-gateway key remain the trust root.
Policy author
π in the policy DSL
salt s (held back)
π
Compiler
π → ConstraintSpec
compiled NFA (JSON)
spec
zkVM guest (SP1)
witness: π, T, trace
public: verdict, violations
L0/L1/L2, delegated
trace
Proof tier
core: no hiding
groth16:
wrapper hiding
(not post-quantum)
proof
Certificate
DSSE/Ed25519 signed
b = H(dom‖n‖T)
C = H(T);
policy_commit = H(s‖π)
violations; proof_mode
n (public input)
certificate
LLM agent
response T (witness)
prompt, decod-
ing parameters
Verifier challenge
nonce n ← 0,1^256
freshness f = |n|
n
Verifier / auditor
checks proof; recomputes b
from delivered T′; checks
ledger inclusion;
no secret state
delivery: T″
inclusion
End user
receives T″; knows n
can recompute b locally
Anchor ledger
append-only;
file or on-chain
32-byte digest
+ timestamp
Divergent delivery (Prop. 1): the certificate
binds the verifier's copy T′, not T″.
public values
witness / secret
verified / anchored
trust boundary